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Simulating climate change impacts on *T. aman* (BR-22) rice yield: a predictive approach using PRECIS and DSSAT models

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Abstract

Climate change has added a new dimension to the uncertainty in rice (*Oryza sativa L.*) yield worldwide and Bangladesh in particular. The present study estimated the potential impacts of climate change on the yield of *T. aman* (Transplanted aman, BR-22) rice variety for 12 representative districts in Bangladesh using DSSAT 4.0 (Decision Support System for Agrotechnology Transfer) model. Available data on soil and typical crop management practice for *T. aman* rice were used in the simulations. The daily weather inputs prerequisite for this model were generated for this study using PRECIS (Predicting Regional Climate Impact Studies) model. The model predicted an average yield of *T. aman* cultivars for selected locations in comparison to the baseline year 2008 are +5.04, +6.06 and –32.43% for the year 2030, 2050 and 2070, respectively. Changes of monthly average temperatures were found to be primarily responsible for the reduction in rice yield. Variability in rainfall pattern over the growing period of *T. aman* rice also affected the yield. Furthermore, model outcomes envisaged that climate change may make rice yield more susceptible to transplanting date, forecasting a significant change in yield as transplanting date is shifted from beginning to the end of July. These outcomes may provide valuable insight into the potential impacts of climate change on rice yield and adequate adaptive measures to mitigate the adverse effect of future climate change in Bangladesh as well as other rice-growing countries.

Introduction

Extreme weather events and climate conditions always make agriculture vulnerable. Regardless of technological advancements such as improved crop varieties, irrigation methods, weather and climate are still key issues in agricultural productivity (Deb *et al.*, 2019). The impact of climate change on food production is one of the major global concerns at the present time. Like as many, Bangladesh, a South Asian country also faces the adverse effects of climate change and in particular, its agricultural sector is highly susceptible due to the climate anomalies (Kabir *et al.*, 2016; Islam and Nursey-Bray, 2017). The economy of the country relies greatly on agricultural production, which is highly climate sensitive. According to Sayeed and Yunus (2018), agriculture has become the single most and largest sector of Bangladesh's economy. This sector currently employs 48% of the rural labour force and accounts for 14.2% of the GDP, of which 70% is contributed alone by rice. However, the agriculture sector is already under threat as a consequence of vulnerable climate issues (Deb *et al.*, 2016; Bocchiola *et al.*, 2019). A minor change in the environment causes a significant change in plant life and plant communities (Murshid, 1987; Islam, 1996). In spite of the fertile land and hardworking people, this disaster-prone country is subjected to food shortage because of high dependency of agricultural production on the weather which is regularly battered by natural disasters (Ahmed, 2017). In 2017/2018, the total food (rice) requirement in Bangladesh was 34.7 million ton while production during this period was 32.65 million ton (Sayeed and Yunus, 2018), making food shortage about 2.05 million ton. Crop–weather linkage using different climatic parameters (for instance, rainfall, temperature, solar radiation and carbon dioxide) is needed to identify potential areas and to characterize the adverse areas with laying emphasis on improvements and improved cultural practices. Bangladesh is seriously lacking in research on climatic variability and associated issues (Paramanik, 1991).

Being the fourth largest rice producer in the world (34.7 million metric tons), the country cultivated an area of about 11.7 million ha (Global Agricultural Information Network, 2018). *Aus, Aman* and *Boro* are the three principal classes of rice, being cultivated in the period from April to July,

August to December and January to May, respectively and dominates country's major cropping pattern. In Bangladesh, the average yield of *Aman* rice (2.19 ton/ha in 2017–2018) is slightly higher than Aus (2.13 ton/ha in 2017–2018), but lower than Boro (3.98 ton/ha in 2017–2018). Bangladesh is critically exposed to weather vagaries that strongly affects the rice yield. Among the climatic parameters, the most important one is rainfall, especially for *T. aman*, which is rain fed (Sayeed and Yunus, 2018). Cultivation of *Aman* rice covers approximately 50.56% of the country's total cultivated land area for rice production (Sayeed and Yunus, 2018). Variation of rainfall patterns, climate change, is likely to affect *T. aman* cultivation. Variability of temperature is also likely to affect the yield of rice (Van Oort and Zwart, 2018). Bangladesh wishes to boost the rice yield to cope with the food shortage for its growing population. However, climate change is an imminent threat to reaching this goal. Therefore, it is important to have a clear concept about the impact of climate change on *T. aman* cultivation, so as to take appropriate adaptation measures.

A number of research has been accomplished so far to assess the impacts of climate variability on rice growth and yield in Bangladesh. Some of these studies are of descriptive type (Faisal and Parveen, 2004; Haque *et al.*, 2016). A few empirical studies were carried out using stochastic regression and crop simulation model to assess the effects of climate anomalies on rice production (Mahmood and Hayes, 1995; Karim *et al.*, 2012; Sarker *et al.*, 2012; Hasan, 2017). Most of these researches were focused either on Boro yield specifically or regardless of country's representative climatic characteristics. Although aman rice covers the largest area of 13 665 acres (48.44%) with a production of 13 190 thousand metric tons rice grain (38 %) (BBS, 2016), studies regarding potential impacts of climate change on *T. aman* rice yield have been very limited. Furthermore, different varieties of a crop are impacted differently by climate change (Sarker *et al.*, 2012). Therefore, a model-based rice production scenario, particularly *T. aman* (BR-22), allied with the future overall climate change scenario considering climatic zones is of immense importance. In these circumstances, this study is carried out to forecast yield of *T. aman* (BR-22) rice under changing climate scenarios in the year of 2030, 2050 and 2070 for 12 major areas of rice farming in Bangladesh. The predictions have been made using the Decision Support System for Agrotechnology Transfer (DSSAT) modelling system.

Methodology

Selection of study area

There are 12 districts of major rice growing areas in different regions of Bangladesh have been selected for the study. These regions were selected on the basis of climatological characteristics (Fig. 1), including north-western, south-central, south-western, southern, south-eastern and north-eastern regions. Among the 12 locations, Rajshahi, Bogra and Dinajpur were selected from the north-western region; Mymensingh and Tangail were selected from central region; Jessore and Satkhira from the south-western region; Barisal and Madaripur from the southern region; Chandpur and Comilla from south-eastern region; and Sylhet district from north-eastern regions (Table 1).

Selection of crop model

The DSSAT Cropping System Model (CSM) has been used for a wide range of applications in agriculture systems. It is proficient in simulating crop growth, development and yield by considering the effects of weather, management, genetics and soil water, carbon and nitrogen. In this study, DSSAT version 4.0 was used for simulations. This tool was chosen for its global use as well as proven performances in impact studies (White *et al.*, 2011). It includes a core driver program (controls timing for each simulation), a land unit module (manages all simulation processes which affect a unit of land), primary modules (discretely simulate several processes that affect the land unit comprising soil processes, soil–plant–atmosphere interface, weather, plant growth and management practices). These components simulate the variations over time in the soil and plants that happened on a single land unit in reply to weather and management practices. The DSSAT-CSM includes models of all crops inside a single set of code. The model component considerably simplifies the simulation of crop rotations as soil courses operate continually and various crops are planted, managed and harvested according to harvesting system statistics given as inputs to the model. A set of data is required for running the DSSAT cropping system model include weather inputs, soil data and crop management data. An inclusive interpretation of DSSAT can be obtained from Jones *et al.* (2003). The model was calibrated and validated for crop yield by comparing the simulated versus observed values. The calibration was performed using the field data collected in 2007 rice season.

Selection of rice variety

The DSSAT model uses a detailed set of crop specific genetic coefficients. The input genetic coefficients are accounted for differences in growth among cultivars. As, crops response to the climate and soil characteristics varies from region to region, therefore it is crucial to find the set of values of the cultivar coefficient for each region (Hasan, 2017). Genetic coefficients allow the model to simulate the performance of different varieties under diverse

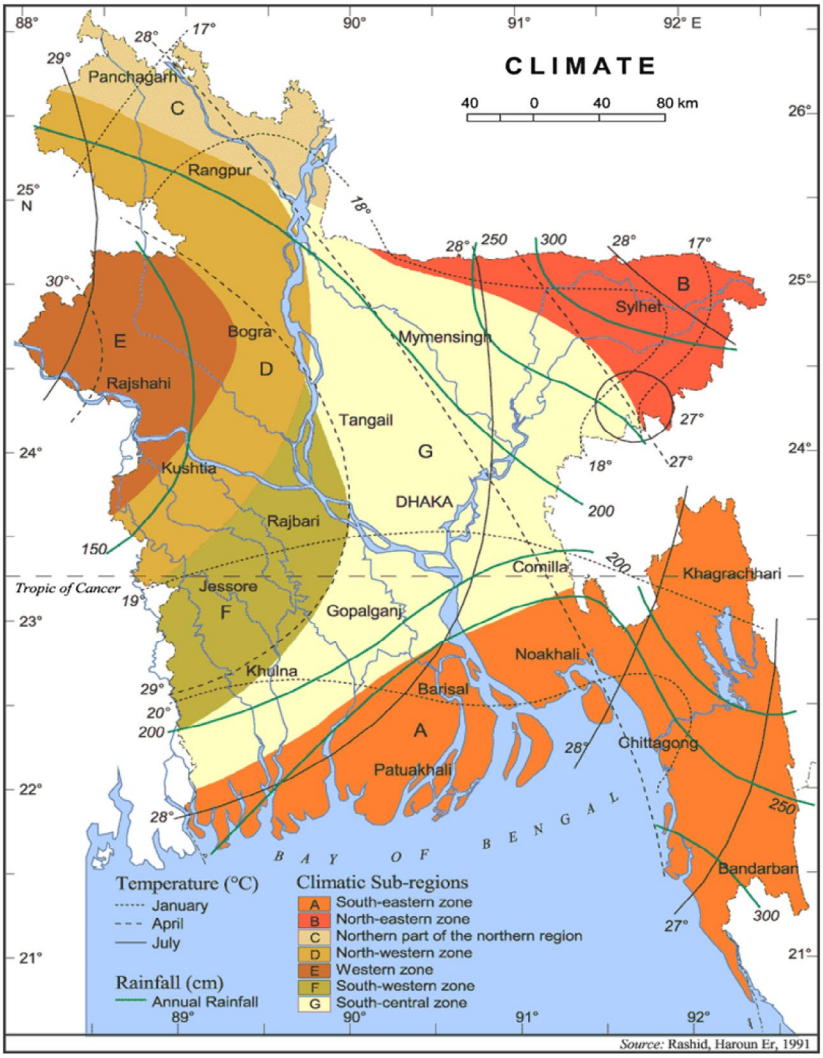


Fig. 1. Climate zone map of Bangladesh (Source: Rashid, 1991).

weather and management conditions. Computation of cultivar coefficients is a fundamental step to define growth and development intending to introduce a new cultivar through crop simulation models (Wallach *et al.*, 2001). Therefore, to obtain sensible outputs from the simulation models, it is essential to have suitable genetic coefficients for the selected cultivars. The *T. aman* rice variety, BR-22 has been selected for this study as it is the country's one of the major cereal crops and the cultivar coefficients of it are available in the model. As the genetic coefficients for BR-22 were not available, the coefficient (Table 2) was obtained from GENCALC (Hunt *et al.*, 1993) of DSSAT v4.0, a computer software program that facilitates estimation of crop genetic coefficients. The calibration of model involved adjusting the values of the genetic coefficients. The best cultivar coefficients were derived (Table 2) based on the location after the iterative process of calibration (Fig. 2).

Soil and crop management inputs

Model inputs related to soil characteristics include upper and lower horizon depth, the percentage of sand, silt and clay, colour, bulk density, organic carbon, permeability, drainage and pH in water. Soil data were collected from the Bangladesh Rice Research Institute (BRRI), Bangladesh Agricultural Research Council (BARI) and Soil Resources Development Institute (SRDI).

Some other data calculated by DSSAT model are a percentage of slope (1%), run-off (moderately low), fertility factor (1), run-off curve number (73), soil albedo (0.13), drainage rate (0.4) and evaporation limit (6.0). Model-calculated some values as like calculate missing values such as a lower limit, a drained upper limit, saturation, bulk density (gm/cm^3), hydraulic conduct (cm/hr) and root growth factor (0–1.0). The crop management data (Table 3) needed

Table 1 Climate variables of twelve districts in different regions of Bangladesh (1981–2010)

Month	Climatic Parameter	North-western			Eastern	Central		South-western		Southern		South-eastern	
		Rajshahi	Dinajpur	Bogra	Sylhet	Mymensingh	Tangail	Jessore	Satkhira	Barishal	Madaripur	Chandpur	Comilla
Jan	T_{max} (°C)	24.1	23.0	24.4	25.6	24.5	23.9	25.5	25.6	25.5	25.1	24.6	25.2
	T_{min} (°C)	12.1	10.4	11.7	12.9	10.6	11.4	11.3	12.1	12.0	13.7	13.5	12.1
	Rain (mm)	8.0	10.2	7.6	6.5	7.1	6.1	13.9	13.2	10.3	7.4	6.0	7.5
Feb	T_{max} (°C)	27.9	26.5	27.5	27.7	27.0	27.5	28.9	28.8	28.5	28.0	27.9	27.8
	T_{min} (°C)	15.9	13.0	14.4	14.9	13.2	14.4	14.7	15.9	15.4	16.3	16.1	15.5
	Rain (mm)	15.1	11.4	13.4	38.4	20.2	27.0	26.0	33.1	26.1	24.7	23.0	22.3
Mar	T_{max} (°C)	33.1	30.9	31.4	30.7	30.6	31.7	33.4	33.0	32.4	31.5	31.7	30.9
	T_{min} (°C)	20.7	17.3	19.0	18.6	17.9	18.9	19.6	20.7	20.5	20.4	20.7	19.7
	Rain (mm)	23.0	12.5	19.6	136.6	39.3	43.8	44.1	39.9	52.4	51.7	62.0	64.8
Apr	T_{max} (°C)	36	32.9	33.5	31.0	31.9	33.9	35.8	35.1	33.5	33.0	33.1	32.4
	T_{min} (°C)	24.4	21.1	22.6	21.1	22.8	22.8	23.6	24.4	23.8	23.6	23.7	22.7
	Rain (mm)	56.5	70.1	75.7	384.6	141.4	110.7	71.6	85.6	103.7	116.7	143.0	141.8
May	T_{max} (°C)	35.1	32.6	33.3	31.2	31.7	33.4	35.4	35.2	33.4	32.8	33.2	32.7
	T_{min} (°C)	25.6	23.2	24.1	22.9	24.4	24.0	25.0	25.6	24.9	24.9	24.8	24.1
	Rain (mm)	136.8	218.3	203.9	563.2	323.0	253.5	182.0	154.2	199.0	228.6	268.0	311.8
Jun	T_{max} (°C)	33.8	32.6	32.8	31.3	31.5	32.7	33.8	33.6	32	31.6	33.2	31.9
	T_{min} (°C)	26.4	25.1	25.8	24.6	25.8	25.6	26.0	26.4	25.8	25.8	25.9	25.4
	Rain (mm)	243.7	366.2	329.3	776.0	402.5	317.6	314.8	294.3	401.7	371.6	371.0	368.5
Jul	T_{max} (°C)	32.5	31.9	32.1	31.5	31.2	31.9	32.6	32.2	31.2	30.8	31.5	31.3
	T_{min} (°C)	26.2	25.7	26.2	25.2	26.1	26.1	26.0	26.2	25.7	25.7	25.9	25.5
	Rain (mm)	303.2	436.0	357.6	768.7	455.1	323.8	325.4	333.8	409.9	378.6	420.0	394.0
Aug	T_{max} (°C)	32.8	32.3	32.5	32.1	31.7	32.2	32.7	32.3	31.4	31.1	31.8	31.9
	T_{min} (°C)	26.2	26.2	26.4	25.3	26.2	26.3	26.0	26.2	25.8	25.8	26.1	25.6
	Rain (mm)	240.6	348.8	279.0	606.2	329.3	243.0	266.7	290.4	342.6	329.1	349.0	291.1
Sep	T_{max} (°C)	32.5	31.7	32.2	31.7	31.4	32.1	32.9	32.3	31.7	31.4	31.8	31.9
	T_{min} (°C)	25.7	25.3	25.7	24.6	25.6	25.7	25.5	25.7	25.4	25.7	25.8	25.2
	Rain (mm)	271.7	362.5	299.9	555.3	329.8	287.8	278.1	299.1	284.4	275.6	285.0	254.1
Oct	T_{max} (°C)	31.7	31.0	31.9	31.4	31.4	31.7	32.5	32.2	31.7	31.5	31.6	31.6
	T_{min} (°C)	23.5	22.3	23.3	22.9	22.9	23.5	23.1	23.5	23.7	24.5	24.3	23.5
	Rain (mm)	115.3	156.8	155.9	189.2	207.8	166.0	134.5	141.4	185.5	157.5	165.0	160.6
Nov	T_{max} (°C)	29.3	28.9	30.2	29.6	29.5	29.4	30.3	30.1	29.8	29.4	29.5	29.7
	T_{min} (°C)	18.4	16.7	18.3	18.7	17.5	18.5	17.8	18.4	18.9	20.5	20.0	18.5
	Rain (mm)	12.6	7.9	10.3	30.3	18.4	26.2	31.0	32.2	48.5	34.9	38.0	34.4
Dec	T_{max} (°C)	25.8	25.3	26.6	26.7	26.1	26.0	26.9	26.9	26.8	26.2	26.2	26.7
	T_{min} (°C)	13.3	12.2	13.7	14.5	12.6	13.5	12.6	13.3	13.6	15.8	15.4	13.4
	Rain (mm)	9.8	8.3	9.9	12.7	9.6	11.6	11.5	7.0	5.9	4.6	7.0	9.5

Source: Khatun *et al.* (2016).

Table 2 Obtained and calibrated cultivar coefficients for *T. aman* (BR22) rice in some locations (Districts) of representative regions of rice production in Bangladesh

Default/calibrated values of diff. locality	Coefficient							
	P1	P2R	P5	P2O	G1	G2	G3	G4
Default value	650.0	110.0	400.0	12.0	60.0	0.025	1.00	1.00
Rajshahi	642.0	120.0	400.0	10.5	70.0	0.027	1.10	1.20
Mymensingh	640.0	110.0	400.0	12.5	65.0	0.025	0.90	1.10
Jessore	645.0	120.0	405.0	13.5	70.0	0.026	1.10	1.20
Barisal	648.0	120.0	402.0	12. 0	62.0	0.025	1.10	1.20
Comilla	655.0	115.0	400.0	10.5	70.0	0.027	1.00	1.10
Sylhet	652.0	120.0	399.0	10.2	63.0	0.027	1.10	1.00

P1: Basic vegetative phase coefficient (expressed as growing degree days [GDD] in °C above a base temperature of 9°C), P2R: Extent in delay of panicle initiation (expressed as GDD in °C), P5: Time period (GDD in °C) from emergence to physiological maturity, P2O: Critical photoperiod at maximum growth rate, G1: Potential spikelet number coefficient, G2: Single grain weight in gm under ideal growing conditions, G3: Tillering coefficient, G4: Temperature tolerance coefficient.

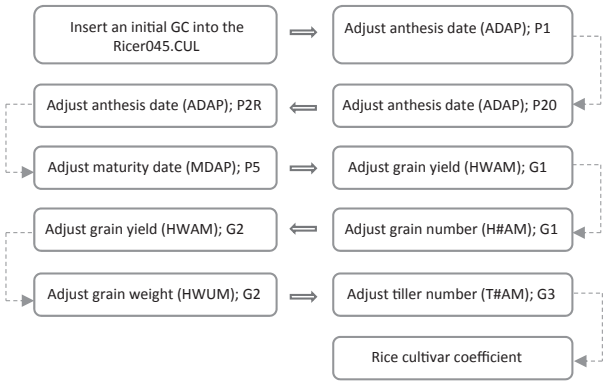


Fig. 2. The sequence of optimizations for calibrating the cultivar coefficients using GENCALC.

Table 3 Crop management data used in the model simulations

Parameter	Input data
Planting method	Transplant
Transplanting date	5,15,20 and 25 July
Planting distribution	Hill
Plant population at the seedling	35 plants per m^2
Plant population at emergence	95 plants per m^2
Row spacing	20 cm
Planting Depth	3 cm
Seedling age	40 days
Plant per Hill	3
Fertilizer (N) application	22 kg/ha
• 18 days after transplanting	40 kg/ha
• 20 days after first application	30 kg/ha
• 22 days after second application	
Application of irrigation	380 mm in 12 applications

Source: BRRI (2006).

for model simulations comprises of planting date, planting density, row spacing, planting depth, irrigation amount and frequency, fertilizer application dates and amounts. Management information was collected from BRRI.

Weather inputs

Climatic data for the period spanning from 1976 to 2005 was collected from 34 stations of Bangladesh Meteorological Department (BMD) for this study. A regional climate model, PRECIS was used to generate the daily weather inputs needed for running the DSSAT crop model. IPCC Special Report on Emissions Scenarios (SRES) A2 of ECHAM4 was used as PRECIS input. It can be mentioned that the PRECIS was simulated in this study with a 50-km parallel resolution for the present climate using baseline lateral boundary conditions (LBCs). The model domain was designated 65–103°E and 6–35°N to cover Bangladesh and its surroundings. After simulation for the year of 2030, 2050 and 2070, the PRECIS outputs were used in the DSSAT model that comprises of daily maximum temperature (T_{max}), daily minimum temperature (T_{min}), daily incoming solar radiation (S_{rad}) and daily precipitation. These parameters were extracted at 12 locations mentioned in Section 2.1. The Fig. 3 illustrates a flow chart of the basic steps taken in the study.

Model performance evaluation

Model performance was evaluated by appropriate calibration and validation with a deviation statistics called root mean square error (RMSE), and a test statistics, called coefficient of determination (R^2). Both these measures are widely used for finding out the accuracy in regression models. RMSE is the absolute fit of the model as it is said to find the standard deviation of variation in data. Its measure goes from zero to infinite the lower the value the better the model. According to Jamieson *et al.* (1991), the simulation is considered excellent with RMSE < 10%, good if 10–20%, fair if 20–30% *and* poor > 30%. On the other hand, R^2 is a measure of how well the regression line represents the data. It indicates the strength of the linear relationship between simulated and observed values, with the value into the range of 0–1, value nearer to 0 poorer the model performance (Hasan and Wyseure, 2018). Furthermore, the model was evaluated with the modelling efficiency (ME). Its range is 1 (optimum value) and it collates modelling uncertainty with experimental variability. RMSE and R^2 and ME are calculated by the equation given herein.

$$\mathrm{RMSE} = \sqrt{\frac{\sum_{i=1}^{n} (S_i - O_i)^2}{n}} \tag{1}$$

$$R^2 = \left(\frac{\sum_{i=1}^{n} (O_i - \bar{O})(S_i - \bar{S})}{\sqrt{\sum_{i=1}^{n} (O_i - \bar{O})^2} \sqrt{\sum_{i=1}^{n} (S_i - \bar{S})^2}} \right)^2 \tag{2}$$

$$\mathrm{ME} = \frac{\left[\sum_{i=1}^{n} (O_i - \bar{O})^2 - \sum_{i=1}^{n} (S_i - O_i)^2 \right]}{\sum_{i=1}^{n} (O_i - \bar{O})^2} \tag{3}$$

where S_i and O_i refers to the simulated and observed values, respectively, for the observation and n is the number of observation within each treatment. $\bar{S}$ and $\bar{O}$ are the mean value of the simulated and observed data, respectively.

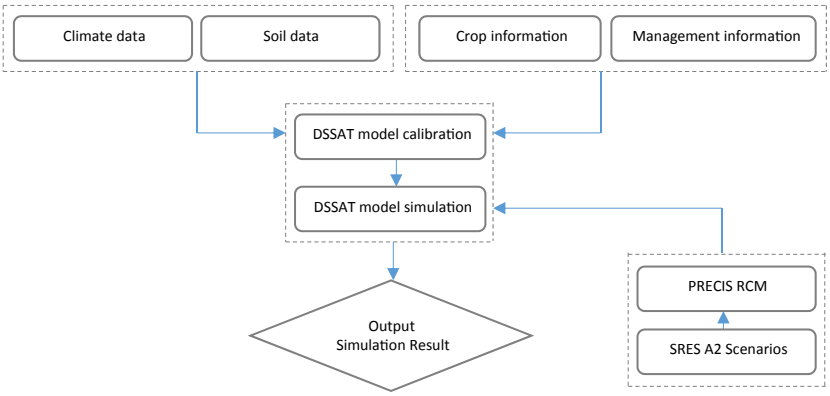


Fig. 3. Flow chart of the basic steps taken in the study.

Results and discussions

Model calibration and validation

Statistical evaluation of the calibration process (7.61 < RMSE < 15.02; 0.81 < R^2 < 0.89 and 0.70 < ME < 0.83) (Table 4) indicated a significant correlation between observed and simulated values. The calibrated model was then validated using independent sets of field data corresponding to 2008. The prediction of crop yields in the validation period was in close agreement with the observed yields (9.90 < RMSE < 14.87; 0.71 < R^2 < 0.82 and 0.70 < ME < 0.83) (Table 4). The calibrated and validated model was then used for scenario analysis.

Table 4 Statistical indices for model performance evaluation of calibration and validation processes for rice yield (*T. aman*, BR-22) simulation at some representative locations (Districts) of rice production in Bangladesh

	Calibration			Validation		
Locality	RMSE	R^2	ME	RMSE	R^2	ME
Rajshahi	15.02	0.81	0.70	26.02	0.71	0.66
Mymensingh	11.23	0.84	0.73	19.23	0.79	0.79
Jessore	10.50	0.87	0.75	17.50	0.72	0.76
Barisal	7.90	0.89	0.80	9.90	0.82	0.84
Comilla	13.08	0.83	0.69	23.08	0.75	0.75
Sylhet	7.61	0.88	0.83	18.61	0.77	0.86

RMSE is the Root Mean Square Error, R^2 is the coefficient of determination and ME is the Modelling Efficiency.

Impact of climate change on rice yield

These predictions have been made using a fixed irrigation application (380 mm in 12 applications), fertilizer dose (92 kg/ha in 3 applications), selected cultivars (BR-22) and CO_2 of 379 ppm (reported for the year 2005 in fourth assessment report of IPCC) for all locations during the year of 2008, 2030, 2050 and 2070. Soil data varied depending on locations; weather data were derived from the PRECIS model. It can be mentioned that 15 July was selected as planting date for all locations in the year of 2008, 2030, 2050 and 2070.

In the future, a significant reduction in rice yield is observed due to the predicted changes in climatic condition (Table 5). The predicted average yield of the selected cultivar for the country's representative locations is +5.04% for the year 2030, +6.06% for 2050 and –32.43% for 2070 in comparison with that in the baseline year 2008 (Table 5). These results are analogous to those indicated by Lal *et al.* (2010); Karim *et al.* (2012); Hasan (2017) in Bangladesh, Soora *et al.* (2013) In India; Van Oort and Zwart (2018) in Africa. Among 12 selected locations, most show yield reduction in the years 2030, 2050 and 2070 in comparison with that in the year 2008. A significant yield reduction occurred at a number of locations; 8.68% at Sylhet (eastern region) in 2030, 17.74% at Rajshahi (north-western region) in 2050 and 40.93% at Barisal (southern region) in 2070 (Table 5). Considerable rise in yield was also predicted; 28.76% at Madaripur (southern region) in 2030 and 30.22% at Satkhira (south-western region) in 2050. It should be mentioned here that an increase in yield to some extent was predicted for some locations, which included north-western region (Bogra) in the years 2030 and 2050, central region (Tangail) in 2050, south-western region (Jessore) in 2030, south-western region in 2050, southern region in 2030 and 2050 and south-eastern region in 2030 and 2050 (Table 5). Figure 4 shows the region-wise prediction of rice yield for BR-22.

Impact of climate change scenarios on *T. aman* yield (CO_2 variable)

For calculating yield of *T. aman* (BR-22) rice under the variability of CO_2 concentrations in years 2008, 2030, 2050 and 2070, 15 July was selected as planting date for all

Table 5 Predicted yield for *T. aman* (BR-22) rice (kg/ha) at selected locations in the year of 2008, 2030, 2050 and 2070

Locality	2008 (379 ppm)	2030 (379 ppm)	2050 (379 ppm)	2070 (379 ppm)	% change in yield for 2030	% change in yield for 2050	% change in yield for 2070
Rajshahi	3534	3428	2907	2311	–2.99	–17.74	–34.60
Bogra	3786	3989	3808	2447	5.36	0.58	–35.36
Dinajpur	3783	3720	3320	2638	–1.66	–12.23	–30.26
Mymensingh	4290	4188	4270	2661	–2.37	–0.46	–37.97
Tangail	4219	4008	4308	2825	–5.00	2.10	–33.04
Jessore	4207	4924	5051	3265	17.04	20.06	–22.39
Satkhira	3414	3339	4446	3113	–2.19	30.22	–8.81
Barisal	3750	4412	4652	2215	17.65	24.05	–40.93
Madaripur	3105	3998	3638	2007	28.76	17.16	–35.36
Chandpur	4583	5035	5216	2802	9.86	13.81	–38.86
Comilla	4743	4966	5018	3174	4.70	5.79	–33.08
Sylhet	4055	3703	3626	2489	–8.68	–10.57	–38.61

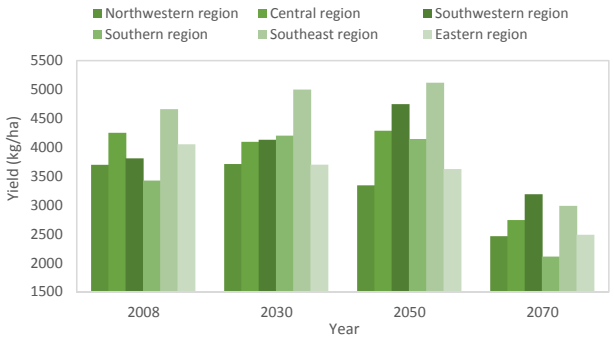


Fig. 4. Region-wise prediction of rice yield on 15 July for BR-22 variety.

locations. The rate of change of atmospheric CO_2 concentration (about 1.9 ppm per year) from 1994 (358 ppm) to 2005 (379 ppm) has been used to set the CO_2 concentrations in the years 2030 (at 427 ppm), 2050 (at 465 ppm) and 2070 (at 503 ppm).

It was observed from the model that at most locations when the CO_2 levels increased to 427, 465 and 503 ppm in 2030, 2050 and 2070, respectively, the yield was increased by 0.15–11.59% (Fig. 5). The predicted average yield changes of *T. aman* (BR-22) for selected locations in comparison with that in the year 2008, are +6.97% for the year 2030, +7.93% for 2050 and -30.13% for 2070. Some regional variations were observed in the predictions (Fig. 5). The maximum yield is increased at Mymensingh (5.80%), Tangail (5.54%) and at Rajshahi (11.59 %) for the *T. aman* cultivar in 2030, 2050 and 2070, respectively.

Therefore, increasing CO_2 concentrations are likely to have a significant positive impact on *T. aman* (BR-22) yield at many locations. The effects of increasing CO_2 on the yield at different regions are given herein (Fig. 5) when the CO_2 is a variable factor.

Sensitivity of yield to climatic parameters

There are many climatic factors that are very much influential in rice production. Some factors have significant roles such as daily maximum and minimum temperature, rainfall, solar radiation and CO_2 which was taken into consideration while model simulation. In order to assess the importance of climatic parameters on predicted rice yield, sensitivity analysis for *T. aman* (BR-22) (Table 6) was accomplished by predicting rice yield in Madaripur district using predicted climatic parameters for the years 2008 and 2070 by changing the value of one parameter, while all other parameters remain the same. The yield of the predicted cultivar will be decreased from 3105 kg/ha in 2008 to 2007 kg/ha in 2070 (Table 5).

Analysis of predicted temperatures shows that average T_{max} during July–November (i.e. *T. aman* season) for 2008 and 2070 are 32.44 and 33.36°C, respectively, and the corresponding values for T_{min} are 23.51 and 26.13°C, respectively (Table 7). If T_{max} and T_{min} of 2070 substitutes T_{max} and T_{min} in 2008, respectively, the predicted yield increase by 11.69% and decreases by 4.70 % (Table 6), respectively. As average T_{max} in 2070 is higher than that in 2008, it should lower the predicted yield, whereas the model prediction shows an increase in yield. The phenomena could be explained by looking at the detail model output for different phases of growth (Table 8). Average T_{max} (when T_{max} in 2070 substitutes T_{max} in 2008) in vegetative phase (Emergence–End Juvenile phase) are lower (within 20–30°C) than that of the vegetative phase in 2008 (Table 8a, b). The high T_{max} (38.6°C) in the vegetative phase is detrimental to rice plant growth and contribute to lowering the yield of rice in 2008.

For all other development phases, T_{max} (when T_{max} in 2070 substitutes T_{max} in 2008) are higher than those in 2008; however, they are within the optimum range (30–35°C). Therefore, it did not affect rice yield and consequently,

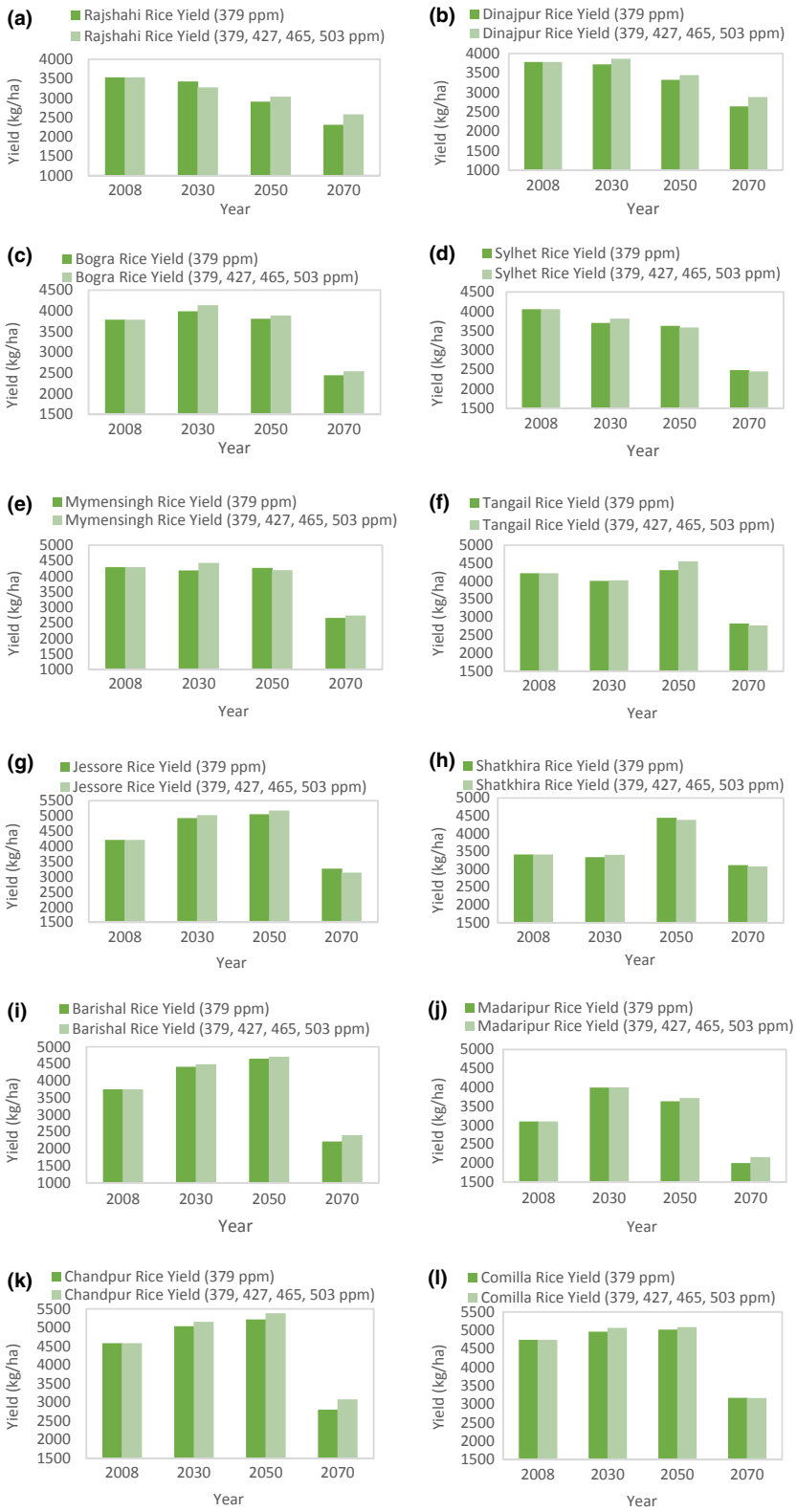


Fig. 5. Predicted yield of *T. aman* (BR-22) rice in northwestern (a, b, c), eastern (d), southwestern (g, h), southern (i, j) and southeastern (k, l) region under different atmospheric carbon dioxide (CO2) concentration.

Table 6 Sensitivity of *T. aman* (BR-22) yield on climatic parameter change at Madaripur

Climatic parameter with year	T_{max}, T_{min} S_{rad}, Rain = 2008	T_{min}, S_{rad}, Rain = 2008 T_{max} = 2070	T_{max}, S_{rad}, Rain = 2008 T_{min} = 2070	T_{max}, T_{min}, Rain = 2008 S_{rad} = 2070	T_{max}, T_{min}, S_{rad} = 2008 Rain = 2070	T_{max}, T_{min}, S_{rad}, Rain = 2070
Rice yield (kg/ha)	3105	3468	2959	2488	2650	2007

Table 7 Monthly average T_{max}, T_{min}, S_{rad} and monthly total rainfall data for Madaripur district in 2008 and 2070 (July–November, i.e. *T. aman* season)

	T_{max} (°C)		T_{min} (°C)		S_{rad} (Mj/m2/day)		Total rainfall (mm)	
Month	2008	2070	2008	2070	2008	2070	2008	2070
July	40.37	34.35	29.24	29.11	21.02	15.15	174.09	385.91
August	31.93	34.59	26.47	28.99	15.04	15.73	352.34	393.31
September	31.99	32.31	25.65	27.51	15.87	11.15	144.02	355.52
October	31.05	35.25	21.41	26.04	17.42	17.88	53.37	52.92
November	26.89	30.3	14.79	19.03	15.28	15.49	3.94	25.28
Average	32.44	33.36	23.51	26.13	16.92	15.08	145.55	242.58

Table 8 Predicted climatic parameters and stresses (from model output) in different development phases of *T. aman* (BR-22) rice in Madaripur

	Environment							Stress (0 = Min, 1 = Max)			
	Average					Cumulative		Water		Nitrogen	
Development phase	Time Span days	Temp Max °C	Temp Min °C	Solar Rad MJ/m2	Photop [day] hr	Rain mm	Evapo-trans mm	Photo synth	Growth	Photo synth	Growth
(a) *Simulation with T_{max}, T_{min}, S_{rad} and Rainfall values of 2008*											
Emergence-End Juvenile	13	38.6	28.8	17.4	13.22	150.3	26.8	0.443	0.475	0.323	0.169
End Juvenile–Panicle Init	19	32.1	26.6	15.0	12.97	208.8	71.1	0.000	0.000	0.471	0.319
Panicle Init–End Lf Grow	32	32.0	26.2	15.5	12.46	265.8	133.7	0.000	0.000	0.465	0.317
End Lf Grth–Beg Grn Fil	9	32.1	25.1	16.0	11.99	29.9	36.5	0.000	0.000	0.000	0.000
Grain Filling Phase	18	30.8	23.2	15.3	11.69	66.2	68.4	0.000	0.000	0.000	0.000
(b) *When T_{max} of 2070 substitutes T_{max} in 2008*											
Emergence–End Juvenile	13	32.6	28.8	17.4	13.37	150.3	29.6	0.241	0.322	0.292	0.168
End Juvenile–Panicle Init	18	34.5	26.6	15.3	13.10	183.9	75.2	0.000	0.000	0.394	0.219
Panicle Init–End Lf Grow	32	32.4	26.2	15.3	12.54	287.6	131.6	0.000	0.000	0.486	0.331
End Lf Grth–Beg Grn Fil	9	35.4	25.1	15.7	12.02	27.2	40.3	0.000	0.000	0.000	0.000
Grain Filling Phase	17	34.2	23.4	15.7	11.69	71.4	67.1	0.000	0.000	0.000	0.000
(c) *When T_{min} in 2070 substitutes T_{min} in 2008*											
Emergence–End Juvenile	13	38.6	28.6	17.4	13.22	150.3	26.8	0.443	0.475	0.324	0.169
End Juvenile–Panicle Init	19	32.1	28.8	15.0	12.97	208.8	71.9	0.000	0.000	0.472	0.320
Panicle Init–End Lf Grow	31	32.0	27.8	15.4	12.47	262.7	131.2	0.000	0.000	0.460	0.315
End Lf Grth–Beg Grn Fil	8	32.2	28.1	15.9	12.03	20.4	33.0	0.000	0.000	0.000	0.000
Grain Filling Phase	16	30.7	27.1	15.3	11.75	78.2	64.9	0.000	0.000	0.000	0.000
(d) *When Rainfall in 2070 substitutes Rainfall in 2008*											
Emergence–End Juvenile	13	38.6	28.8	17.4	13.22	221.4	59.8	0.300	0.354	0.321	0.166
End Juvenile–Panicle Init	17	32.1	26.5	15.1	12.99	152.2	72.1	0.000	0.000	0.446	0.299
Panicle Init–End Lf Grow	32	32.0	26.3	15.3	12.50	572.2	129.4	0.000	0.000	0.573	0.405
End Lf Grth–Beg Grn Fil	9	32.0	25.1	15.9	12.04	34.3	36.2	0.000	0.000	0.183	0.026
Grain Filling Phase	18	30.9	23.5	15.6	11.73	62.1	69.1	0.000	0.000	0.000	0.000
(e) *When S_{rad} in 2070 substitutes S_{rad} in 2008*											
Emergence–End Juvenile	13	38.6	28.8	8.3	13.22	150.3	29.7	0.195	0.258	0.324	0.167
End Juvenile–Panicle Init	18	32.1	26.6	17.2	12.98	183.9	78.7	0.000	0.000	0.599	0.421
Panicle Init–End Lf Grow	32	32.0	26.2	10.1	12.48	287.6	86.6	0.000	0.000	0.241	0.172
End Lf Grth–Beg Grn Fil	9	32.0	25.1	18.0	12.02	27.2	39.7	0.000	0.000	0.000	0.000
Grain Filling Phase	18	31.0	23.4	16.4	11.71	71.4	75.4	0.000	0.000	0.000	0.000

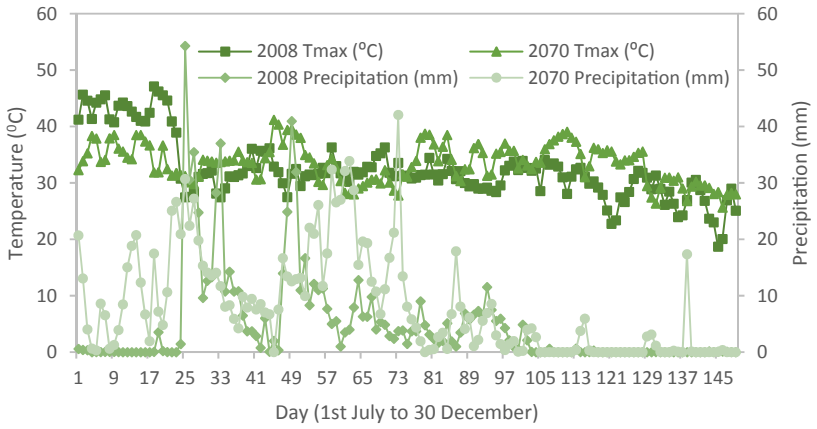


Fig. 6. Variation of T_{max} and rainfall from July to November in 2008 and 2070 at Madaripur districts.

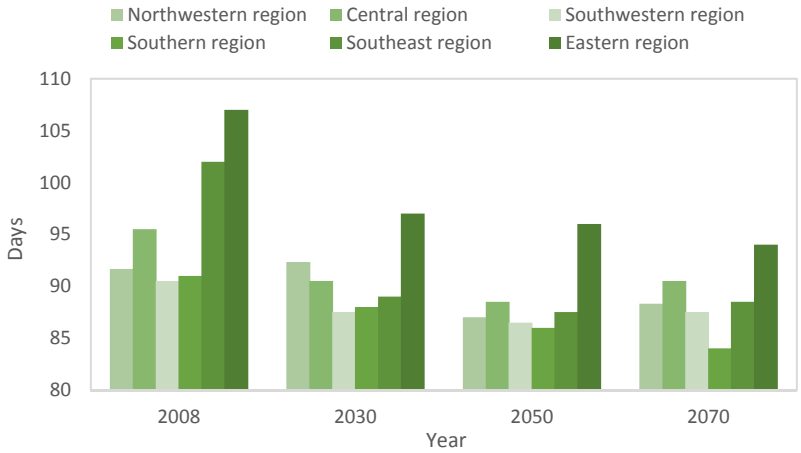


Fig. 7. Effects of climate change on the physiological maturity of *T. aman* (BR-22) in different regions (Planting date: 15 July).

predicted yield becomes higher. Average T_{min} for 2070 in vegetative, reproductive and ripening phase are higher than the corresponding values of T_{min} in 2008 (Table 8a, c), which contribute to decrease in rice yield.

On the other hand, predicted average rainfall during July–November (i.e. *T. aman* season) for 2008 and 2070 are 145.6 and 242.6 mm, respectively, and the corresponding values for average S_{rad} are 16.92 and 15.08 MJ/m^2/day, respectively (Table 7). If rainfall and solar radiation S_{rad} of 2070 substitutes rainfall and S_{rad} in 2008, the predicted yield decreases by 14.65 and 19.87 % (Table 6), respectively, for BR-22 variety. It happens because excessive rainfall is predicted during reproductive phase (i.e. Panicle Initiation–End Leaf Growth phase), again slightly higher rainfall occurred during vegetative phase (i.e. Emergence–End Juvenile phase), reproductive–ripening phase (i.e. End Leaf Growth–Beginning Grain Filling phase) and slightly lower rainfall occurred during vegetative–reproductive phase (i.e. End Juvenile–Panicle Initiation phase) and ripening phase (i.e. Grain Filling phase) in 2070 compared to those in 2008 (8a, d). Similarly, the changes in yield predicted by the DSSAT model could be explained for other locations as well. Figure 6 shows that variation of T_{max} and rainfall during July to November in 2008 and 2070 at Madaripur districts.

S_{rad} values for the year 2070 are lower than those of 2008 except vegetative–reproductive phase (i.e. End Juvenile–Panicle initiation), reproductive–ripening phase (i.e. End Leaf Growth–Beginning Grain Filling) and ripening phase (i.e. Grain Filling) (Table 8a, e). This change in solar radiation appeared to be responsible for the forecasted decrease in yield.

Effect of climate change on physiological maturity

The predicted duration of physiological maturity of BR-22 for 12 different rice growing areas decreased significantly

Table 9 Predicted yield (kg/ha) of *T. aman* (BR-22) in July for four different planting dates

	Date in the year															
	2008				2030				2050				2070			
Locality	5th	15th	20th	25th	5th	15th	20th	25th	5th	15th	20th	25th	5th	15th	20th	25th
Rajshahi	2740	3534	2383	2743	3036	3428	3860	4023	3387	2907	3291	3179	2253	2311	2784	2469
Bogra	3377	3786	3403	3067	3756	3989	4011	4078	3967	3808	3937	3346	2480	2447	2644	2385
Dinajpur	3925	3783	3657	3149	3653	3720	3828	4128	3631	3320	3321	3433	2671	2638	2602	2398
Mymensingh	3847	4290	3598	3258	4066	4188	4486	4495	4655	4270	4197	4111	2769	2661	2829	2846
Tangail	4095	4219	3595	3396	3508	4008	4112	4147	3695	4308	4190	3527	2697	2825	2929	2812
Jessore	3107	4207	3758	3898	4544	4924	5148	5141	5275	5051	4894	4251	3209	3265	3269	3328
Satkhira	2656	3414	3041	3167	3212	3339	4634	4827	4071	4446	4361	3955	2780	3113	3237	3112
Barisal	3432	3750	3648	2978	3948	4412	4818	4483	4594	4652	4452	3455	2646	2215	2595	2702
Madaripur	2807	3105	2395	3144	2517	3998	4299	4418	3821	3638	3984	3454	2459	2007	2438	2606
Chandpur	4438	4583	4264	3785	4541	5035	5150	4880	5334	5216	5017	3985	3067	2802	3165	3366
Comilla	6152	4743	4423	4931	4611	4966	5061	5127	5294	5018	4729	4235	2968	3174	3268	3185
Sylhet	3891	4055	3404	3145	3876	3703	3640	4385	4001	3626	4052	3809	2281	2489	2484	2177

due to changes in the climatic scenario. For instance, physiological maturity (in days) for Comilla varied from 112 days in 2008 to 90 days in 2070. It should be mentioned that the DSSAT model reckons physiological maturity from the time of 'Emergence to End Juvenile' period. It takes about 10–12 days to come to this stage after transplanting.

Some regional variation is observed in model predictions for *T. aman* (BR-22), physiological maturity decreased the most for south-eastern region (by 25 days) and the least for the south-western region (by 6 days) between the years 2008 and 2070 (Fig. 7). Changes in temperature and solar radiation are important factors, affecting the physiological maturity of rice. Physiological maturity also depends on planting date.

Effect of planting date on rice yield

The model predictions show that the predicted yield of *T. aman* (BR-22) could vary significantly with planting date (Table 9). There is no clear trend was observed for optimum planting date and year. For instance, 15 July appears to be the optimum planting date for 2008, 25 July for 2030, 5 July for 2050 and 20 July for 2070 are the best among four selected planting date. However, some regional variation for optimum planting date is also observed (Table 9). For instance, in 2008, 15 July was the most productive time for the north-western, central, south-western, southern, eastern regions and 5 July was the best for south-eastern region.

Conclusions

(1) This research work renders predicted yield of *T. aman* (BR-22) under changing climate scenarios in years 2030, 2050 and 2070 in 12 representative areas for rice cultivation in Bangladesh.

(2) The forecasts have been made using the DSSAT modelling system.

(3) The outcomes of this study revealed that a significant reduction in T. aman rice yields in the future is predicted due to changes in climatic condition.

(4) The predicted average yield of T. aman (BR-22) cultivars for the selected locations in comparison to the baseline year 2008 are +5.04, +6.06 and –32.43% for the year 2030, 2050 and 2070, respectively.

(5) The model predicted significant yield reduction at a number of locations; 8.68% at Sylhet in 2030, 17.74% at Rajshahi in 2050 and 40.93% at Barisal in 2070.

(6) However, an increase in yield was also predicted at Madaripur (28.76%) in 2030 and at Satkhira (30.22%) in 2050.

(7) Projected future changes in yield of this cultivar could be explained by changes in the climatic parameters in different development phases of plant growth.

(8) Increases in mean monthly temperature have been found to be mostly influential on the predicted changes in *T. aman* rice yield.

(9) Variations in rainfall pattern over the growing period of *T. aman* could also affect the yield.

(10) Increasing atmospheric CO_2 concentration is likely to have some beneficial influence on rice yield, which could offset some of the adverse effects of other climatic parameters on rice yield.

(11) The optimum planting date (for maximizing yield) appears to be a shift from the middle of July in 2008 to end of July in 2070, except 5 July for 2050 of *T. aman* (BR-22) rice variety.

(12) The use of other crop simulation models (i.e. APSIM, WOFOST, EPIC) together with dynamically downscaled

climatic inputs for rice production in this region under various treatments and management conditions (irrigation, fertilizer, chemical applications, etc.) needs to be addressed for the enhancement of the reliability of the results.

(13) Overall, it can be said that the DSSAT model is very competent in simulating the potential effects of climate change on rice yield.

(14) Therefore, it can be considered a useful tool that aid in making an effective decision by means of assessing crop management strategies as well as crop responses to climate change.

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Competing interests

The authors of this article declare that they have no potential conflict of interests.

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